

Math 115A Spring 2026 Discussion 19 Handout

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1 Introduction

Today we'll discuss some tips to prepare for the final, and work on a few problems relating to inner product spaces.

2 Final Exam Advice

First, some advice for studying over the next week-and-a-half.

1. Do problems! This is the only way to learn the material. And don't just solve them in your head - sit down and write the solution the same way you would on an exam, or type them. Often, the process of writing a proof will reveal logical errors or subtleties you missed while just thinking about the problem - this is important! And even if it doesn't, you should practice writing proofs for an audience, rather than for your own notes. Try working with a classmate or friend - show them your proof for a problem they haven't thought about, and see if they can follow it. You can also email your solutions to me, or ask AI to check them. (If you go the AI-route, be warned that it might not catch under-justified statements - in this class, we really want you to justify everything, but in more advanced classes one can get away with skipping over sufficiently "trivial" details).
2. Know the definitions! Drill them with flash-cards if you have to. When you're doing a practice problem, the first thing you should do is write down the definition of every term in the problem. If you can't remember the definition, it's OK to check your notes, but after doing this enough you'll find you've memorized everything. The definitions are the most important part of any proof-based math course; without them you won't even be able to start thinking of proofs.
3. Once you know all the definitions - meaning, you should be able to recall them immediately - move on to the major theorems (anything labeled "theorem" in the class notes, but I'd include the Steinitz Exchange Lemma in here too). The statements here are most important, but if you have time it's good to learn the proofs as well.

4. The quality of your practice problems are more important than the quantity. It's better to spend 2 hours struggling with a single problem that you solve yourself than to get through 15 problems by peeking at the solutions. Don't get discouraged if it takes you a long time to spot an "easy" solution! Often, the value of exercises is that they give you a feel for the ideas of the course, and the process of solving them is much more important than the ultimate solution.
5. Do a few concrete problems! You should understand how to compute the eigenvalues and eigenvectors of small (2×2 or well-chosen 3×3 matrices), for example. You should probably do at least one explicit computation of a change-of-basis matrix. Pick three vectors in $\mathbb{R}^3$ and use Gram-Schmidt to obtain an orthonormal basis. Do an explicit example of the Steinitz Exchange Lemma. Et cetera. For every abstract result we've proven, you should work out a concrete example to see what the theorem actually means. This can be particularly helpful for results you don't understand!
6. Similarly, you should get familiar with some common counterexamples (particularly for true or false questions). These include (but are not necessarily limited to):
 - (a) "Zero objects." If a problem asks about a general vector, see if it's true of the zero vector. If a problem asks about a general vector space, see if it's true of the zero subspace. If a problem asks about a general linear map, see if it's true of the zero map.
 - (b) Likewise, "trivial examples." If a problem asks about subspaces $W \subseteq V$, check whether it's true for $W = V$. If a problem asks about linear maps, test it out on the identity map (or scalar multiples of the identity map).
 - (c) If you see a statement about all vector spaces, rather than just finite-dimensional ones, you should be suspect that it might not hold in the infinite-dimensional case. The go-to example here should be the space $\mathbb{R}^{\mathbb{N}}$ of sequences of real numbers; the "translation maps" that shift a sequence to the left or to the right can be particularly useful, but you might need to construct a more elaborate map to suit the scenario.
 - (d) If you see a statement about eigenvalues, characteristic polynomials, and diagonalizability, it's helpful to know some non-diagonalizable maps: maps given by matrices of the form $\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$, and in particular $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, are my favorite examples.

And now, some problem-solving advice for the big day.

1. You should always start by writing down all the relevant definitions. This will help you keep track of the information you're given at the start, and give you a better idea of what you're trying to prove.

2. Think about whether you should break the problem into parts. There are several problems that you should always break up in about the same way. For example:
 - (a) If you are asked to prove “ p if and only if q ,” you should prove “if p , then q ” and “if q , then p .”
 - (b) If you’re given $A, B \subset V$ subspaces of a vector space, and want to show that $A = B$, you should show $A \subseteq B$ and $B \subseteq A$.
 - (c) If you want to prove that a function $f : A \rightarrow B$ is bijective, you should prove that f is injective and that f is surjective.
 - (d) If you want to prove that a set $\mathcal{B}$ is a basis for a vector space V , you should show that $\mathcal{B}$ is linearly independent and spans V .
 - (e) Sometimes, the easiest way to prove that $\dim(V) = n$ is to show that $\dim(V) \leq n$ and $\dim(V) \geq n$. That is, find a spanning set of size n , and then a linearly independent set of size n - they don’t need to be the same set! Of course, you can also just show that a given set of size n is a basis for V .
 - (f) To show that a function $f : V \rightarrow W$ is a vector space isomorphism, you should prove that f is linear and that f is bijective. Or, prove that f is linear, that $\dim(V) = \dim(W)$, and that f is either injective or surjective.
3. After potentially breaking your proof into smaller pieces, you should look at each part and figure out what the first and last lines of the proof should look like. Often these will follow a formula. For instance:
 - (a) To show that a set $\{v_1, \dots, v_n\}$ is linearly independent, you should always start by choosing a linear combination $\sum_i a_i v_i = 0$ and end by showing that $a_i = 0$ for all i .
 - (b) To show that a set $\{v_1, \dots, v_n\}$ spans V , you should start by picking an element $y \in V$ and end with a linear combination $\sum_i a_i v_i = y$.
 - (c) To show that a linear map $f : V \rightarrow W$ is injective, you should start by considering $v \in V$ such that $f(v) = 0$, and end by showing that $v = 0$. If you don’t know that f is linear, then you should start by considering $v, w \in V$ with $f(v) = f(w)$, and end by showing that $v = w$.
 - (d) To show that a set $\{v_1, \dots, v_n\} \subset V$ is linearly *dependent*, there’s no set way to start, but you’ll always end by finding a non-trivial linear combination $\sum_i a_i v_i = 0$.

Usually, if you’re trying to prove something of the form “if p , then q ,” or “object X has property q ,” the last line will resemble the definition of q . The first line will typically use the definition of p or X . So you should always start by writing these definitions down!

4. One very popular technique is proof by contraposition or by contradiction: to show “ p implies q ,” you can instead show either: (1) that “not q implies not p ” or (2) that “not q ” implies some falsehood. If you can’t figure out how to approach a problem, it’s often worth exploring what happens when you assume the conclusion is false. Even if this isn’t the right way to approach the problem, I find that considering why the result might *not* be true helps me come up with a proof that it is.
5. If you’re still not sure what to do, see what happens in an easy concrete case, like $\mathbb{R}^2$ or $\mathbb{R}^3$.
6. If a proof comes in multiple parts, you should assume that these parts are grouped together for a reason. Sometimes the statements of parts (b) or (c) will help you figure out part (a); on the other hand, you’ll often want to use the results of earlier parts of a problem to solve later parts. Just make sure not to use later parts to solve earlier parts!

Inner products, orthonormality, and duality

First, here’s a problem we didn’t get to last week.

Exercise:

- (a) Let $f : W \rightarrow V$ be an injective linear map of vector spaces over $\mathbb{R}$ and suppose V has an inner product $\langle \cdot, \cdot \rangle_V$. Show that the function $\langle \cdot, \cdot \rangle_f : W \times W \rightarrow \mathbb{R}$ given by $\langle w_1, w_2 \rangle_f = \langle f(w_1), f(w_2) \rangle_V$ is an inner product on W . We sometimes call $\langle \cdot, \cdot \rangle_f$ the **pullback** inner product.
- (b) In particular, the result of part (a) holds when f is an isomorphism. If $\mathcal{B} = \{v_1, \dots, v_n\}$ is an orthonormal basis for $(V, \langle \cdot, \cdot \rangle_V)$, show that $f^{-1}(\mathcal{B}) = \{f^{-1}(v_1), \dots, f^{-1}(v_n)\}$ is an orthonormal basis for $(W, \langle \cdot, \cdot \rangle_f)$.
- (c) Now, let W be any finite-dimensional vector space with basis $\mathcal{A} = \{w_1, \dots, w_n\}$. Show that there exists an inner product $\langle \cdot, \cdot \rangle_W : W \times W \rightarrow \mathbb{R}$ such that $\mathcal{A}$ is an orthonormal basis for $(W, \langle \cdot, \cdot \rangle_W)$. [Hint: use the result of part (b) with a good choice of linear map f].

It’s a good exercise to check that the above works over $\mathbb{C}$ as well, but I didn’t want the more involved definition of a complex inner product to get distracting.

Exercise: As a straightforward computational exercise: Consider $\mathbb{R}^3$ with the standard inner product. Consider $\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \\ 7 \end{pmatrix}, \begin{pmatrix} 4 \\ 8 \\ 9 \end{pmatrix} \right\}$. Following the steps of the Gram-Schmidt Orthogonalization proof, obtain an orthonormal basis. As a follow up exercise, construct a different inner product on $\mathbb{R}^3$

Once you feel comfortable with concrete examples of Gram-Schmidt orthogonalization, try to prove the general statement on your own!

Exercise: Let $(V, \langle \cdot, \cdot \rangle)$ be a finite-dimensional inner product space and $W \subset V$ a subspace. Show that $(W^\perp)^\perp = W$. (Hint: you may want to start by picking a convenient orthonormal basis for V).

Finally, here are a couple problems about duality. Recall that if V is a vector space over the field $\mathbb{F}$, then the dual of V , denoted V^* , is $\mathcal{L}(V, \mathbb{F})$.

Exercise: Let $(V, \langle \cdot, \cdot \rangle)$ be a finite-dimensional inner product space. Show that $V \cong V^*$ by constructing an isomorphism. (Hint: again, you may want to start by picking an orthonormal basis for V).

Remark: The isomorphism $V \rightarrow V^*$ in the above problem depends on the choice of inner product on V . If you don't believe me, give V a different inner product and see whether you get the same map $V \rightarrow V^*$!

Exercise: Let V be a finite-dimensional vector space over a field $\mathbb{F}$. Let V^{**} denote the “double-dual” of V ; that is, $V^{**} = \mathcal{L}(V^*, \mathbb{F})$.

1. Without choosing a basis for V , find an isomorphism $\Psi_V : V \rightarrow V^{**}$. (Hint: try to describe what V^* and V^{**} represent “in words” first). You should get a “recipe” for Ψ_V that makes sense when V is replaced with any other $\mathbb{F}$ -vector space W .
2. Suppose W is another vector space over $\mathbb{F}$ and let $f : V \rightarrow W$ be another $\mathbb{F}$ -linear map. Show that f induces a linear map $f^* : W^* \rightarrow V^*$ given by $f^*\phi(v) = \phi(f(v))$. [Hint: start by figuring out what vector spaces ϕ and v live in].
3. It follows that f^* induces a map $f^{**} : V^{**} \rightarrow W^{**}$. Taking Ψ_V and Ψ_W to be the maps $V \rightarrow V^{**}$ and $W \rightarrow W^{**}$ from part (a), show that $f^{**} \circ \Psi_V = \Psi_W \circ f$. [Hint: Start with a vector $v \in V$, and show that $f^{**}(\Psi_V(v)) = \Psi_W(f(v))$. It might help to figure out what sort of object $f^{**}(\Psi_V(v)) = \Psi_W(f(v))$ is]. This amounts to showing that the below diagram “commutes:”

$$\begin{array}{ccc} V & \xrightarrow{\Psi_V} & V^{**} \\ f \downarrow & & \downarrow f^{**} \\ W & \xrightarrow{\Psi_W} & W^{**} \end{array}$$